

system budgets. EMI/EMC screening will be conducted to confirm that CDH subsystems do not interfere with other avionics or payloads and are immune to nearby signals.

The table below summarizes verification methods used for each CDH requirement category:

Requirement Category	Verification Method
Command reception and execution	Functional test, software simulation
Data logging and ECC integrity	Write/read tests, fault injection
CAN/SpaceWire communication	Bus emulation, traffic load simulation
Communications protocol (CCSDS)	Emulator-based packet transmission
Safe mode and FDIR recovery	Fault injection, reboot recovery test
Mass and power limits	Inspection, measurement
Environmental survivability	Thermal/vibration testing, EMI analysis

All results will be logged and included in the CDH Verification Report submitted with the team's final Preliminary Design Review (PDR) package. Successful verification ensures the CDH is fully compliant with design goals, mission objectives, and system-level integration requirements.

2.1.4. Thermal Control Subsystem Overview

To withstand Venus's harsh environments, two of the primary challenges are withstanding the high wind speeds and the significantly high temperatures. Nevertheless, since this is an entirely aerial mission the insanely high surface temperatures are neglected. Moreover, space-grade mechanical structural design and electrical/science components provide an operating temperature closer to the environmental temperature. The mission descends to about a minimum altitude of about 35 km from a maximum altitude of 50 km. This introduces the hottest condition at

the minimum altitude at a temperature of about 400 K and the coldest condition at the maximum altitude at a temperature of about 350 K. Note, as discussed earlier, since the system was designed with space-grade components fit for Venus's conditions their operating temperature ranges from 350 K - 400 K. However, for them to optimally perform and mitigate risk of damage or failure, a safer operating temperature of 370 K was chosen for both system thermal integrity and thermal control system manufacturing feasibility within mission constraints.

The primary modes of heat transfer were investigated to be radiation and convection, with convection being the challenge in this particular mission. Internal conduction was neglected by relying on the mechanical/electrical structural integrity and assuming it is factored in as internal load, and since this is an aerial mission there is no conduction with the environment. Solar flux is considered however, it was found to have not been as significant since Venus's atmosphere blocks most of it. Radiation was found to be considerable but severely insignificant when compared to the effects of convection. Nevertheless, effects of radiation were handled to bring down the magnitude of its effect by 100x with simple thermal management to aid the thermal control subsystem in handling the significant effects from convection. Lastly, perhaps the greatest challenge was convection because of Venus's not only high temperature but also high winds.

Stefan-Boltzmann Law described by

$$Q_{\text{radiation}} = \varepsilon \sigma A (T_s^4 - T_{\text{environment}}^4)$$

where ε is the emissivity, σ is the stefan-boltzmann constant, A is the effective surface area, and T stands for temperature values was used to compute radiation effects on the aerobot system and Newton's Law of Cooling described by

$$Q_{\text{convection}} = hA(T_s - T_{\text{environment}})$$

where h is the convection heat transfer coefficient, A is the effective surface area, and T stands for temperature was used to compute convection effects on the Aerobot system where the convection heat transfer coefficient was determined using the Churchill-Bernstein approach.

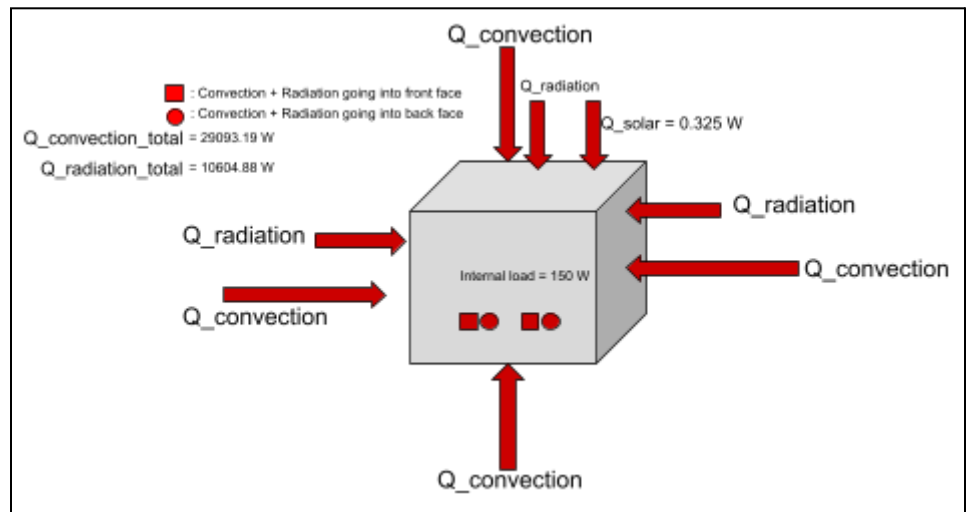


Figure 14. Heat flow map of the hottest condition

As illustrated by the heat flow map in the hottest temperature extreme condition assuming average wind speed of 90 m/s, the major heat transfer effect is by convection as expected at about 30 kW of thermal energy in and thus 30 kW of cooling is required.

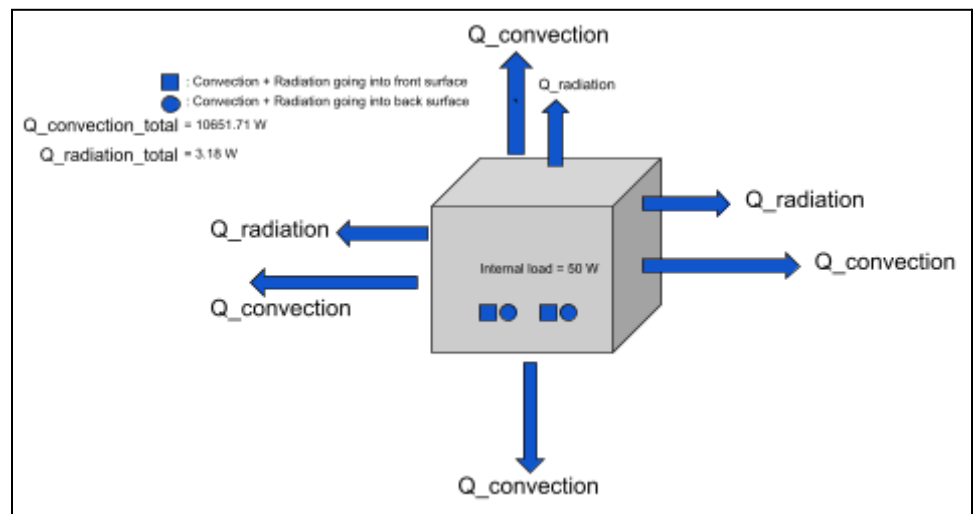


Figure 15. Heat Flow Map of the Coldest Condition

As illustrated by the heat flow map in the coldest temperature extreme condition assuming average wind speed of 90 m/s, the major heat transfer effect is by convection as expected at 11 kW thermal energy leaving and thus 11 kW of heating is required.

2.1.4.1. Thermal Control Subsystem Requirements

The subsystem requirements were formulated to ensure thermal management optimal functionality. The purpose of the thermal control subsystem is to maintain the Aerobot system at a safe operating temperature derived from the mission environment and Aerobot system component thermal needs to operate at maximum performance levels. The mission environment requires thermal management to handle high temperatures at a range of 350 K to 400 K, and a system temperature of 370 K was decided as the safe operating temperature considering mechanical, electrical, and science instrumentation thermal capabilities and needs. Design of thermal control system includes that of chemical coating, white painted surfaces, and an ACT radiator. The chemical coating and white painted surfaces are required to manage relatively light thermal loads including that of radiation, internal loads, and solar flux. The ACT radiator is required to manage significant convection thermal loads from the high speed winds by having the capability to provide at least the required 30 kW cooling and 11 kW heating. All the above discussed requirements can be analytically assessed through simulation, however since a Venus-like environment can be formulated as a testing environment, testing verification method was given priority since it is the most reliable.

Table: Thermal Control subsystem Requirements Table

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Relevant Subsystem	Req met?
TH 2.1	The Thermal subsystem shall maintain a system equilibrium temperature of 370 K.	The Aerobot system internal temperature needs to be about 370 K to keep all science instrumentation and electrical components safe.	TH-1	-	Test	All	Met

TH 2.2	All Thermal subsystem components shall optimally perform in the temperature range of 350 K - 400 K.	All Thermal subsystem components need to be functioning optimally in the coldest and the hottest conditions to successfully regulate the Aerobot system's temperature.	TH-1	-	Test	All	Met
TH 2.3	All Aerobot system MLI coating and white paint surfaces shall perform optimally.	It is imperative of thermal management to have effective MLI coating as it negates the solar flux and the white surfaces maximize heat transfer out of the Aerobot system.	TH-1	-	Test	All	Met
TH 2.4	The Aerobot system shall be equipped with an optimally functioning ACT radiator to provide with the needed amount of 30 kW cooling in the hottest and 11 kW of heating in the coldest conditions.	It is imperative of thermal management to have a successfully functioning quality ACT radiator to bring up the temperature of the Aerobot system to maintain target equilibrium temperature of 370 K.	TH-1	-	Test	All	Met

2.1.4.2. Thermal Control Sub-Assembly Overview

As discussed earlier, convection from the winds is the largest contributor of heat transfer in this mission because of Venus’s high winds. Nevertheless, effects from radiation and solar flux are managed by simple but very effective thermal control systems including that of painting the side surfaces of the Aerobot system with white paint and the top surface of the Aerobot system with chemical coating. The SilcoNert® 1040 Coating can easily withstand harsh Venus temperatures as it is rated for temperatures up to 450 K. With an average emissivity of 0.008, the chemical coating reduced the total radiative effects of 466.64 W to 9.33 W. Although it is primarily designed for chemical inertness properties, its thermal control abilities are useful considering the 0.008 emissivity.

Next, the convective effects are handled using a variable-view-factor two-phase radiator panel from Advance Cooling Technologies (ACT). Firstly, the radiator is passive so it does not require any power. It also does not need any control system or temperature input as it operates autonomously based on its internal pressure vapor pressure. Changing its geometry based on its internal vapor pressure, it can control how much thermal energy is radiated out of the system. When temperature drops the internal vapor pressure drops and the radiator closes up reducing heat loss from the system helping to trap heat and retain system temperature. On the other hand when temperature increases the internal vapor pressure increases and the radiator opens up to increase surface area and reject heat. Moreover, this radiator from ACT is also an efficient system as it is two-phase. Compared to a single phase radiator utilizing conduction/convection, a two-phase radiator much more efficiently rejects heat by utilizing a working fluid inside that rejects heat by changing the phase of the fluid to vapor. This helps absorb a larger amount of heat and also rejects heat faster when the liquid condenses after they reach the fins on the surfaces.

Table: Thermal Control Subsystem component details

	ACT Variable-View-Factor Radiator Panel	SilcoTek SilcoNert 1040
Weight (kg)	4	Negligible
Dimensions (m)	.8 x .3 x .3	N/A
Max Power Drawn (W)	Passive	No Power needed
Quantity	1	As much needed for surface area of 1 m ²
TRL	5	5

2.1.4.3. Thermal Control Subsystem Recovery and Redundancy Plans

The ACT variable radiator has a passive simple mechanical design and thus the likelihood of functional failure is reduced. Nevertheless during the mission, cooling is significantly more important than retaining heat and thus in case of any failure detected the radiator will default to an open state to maximize heat transfer out of the Aerobot system to effectively cool it down. Moreover, the radiator can still perform with a lower surface area but still keep the Aerobot system safe noting the thermal ratings of the Aerobot system is capable of withstanding a range of temperatures around the chosen safe system temperature.

Besides the ACT variable radiator, the SilcoNert

1040 chemical coating also aid in thermal control even if in much less of a capacity. Thus, in the event of the ACT radiator underperforming the chemical coating can ensure thermal control. Nevertheless, localized surface degradation can occur and thus multiple overlapping layers of coating mitigate the likelihood of this damage.

2.1.4.4. Thermal Control Subsystem Manufacturing and Procurement Plans

SilcoTek Corporation and Advanced Cooling Technologies (ACT) are both exclusive vendors of both the thermal components. Nevertheless, they are both specialized vendors fit for both the specialized components. SilcoTek Corporation has unique expertise of the chemical coating needed and they also do the coating in-house mitigating quality control and lead time contingencies. Secondly, Advanced Cooling Technologies Inc. has experience in space-grade component manufacturing (NASA collaboration).

Component: SilcoTek SilcoNert 1040

Supplier: SilcoTek Corporation

Justification: They do the coating on the metal in-house ensuring quality control

Specialization: Unique expertise in chemical coating of this kind

Lead Time:

- Priority coating: 5 or more business days
- Expedited coating: 3-4 business days

Component: ACT Variable-View-Factor Radiator Panel

Supplier: Advanced Cooling Technologies, Inc. (ACT)

Justification: Exclusive vendor of this Variable-View-Factor radiator

Specialization: Experience in space-grade radiators (NASA collaborations)

Lead Time: It is not publicly disclosed so it is advised to contact directly

- Estimation can be made to be about 6 months for a custom design
- They might have existing designs which might decrease lead time

2.1.4.5. Thermal Control Subsystem Verification Plans

Table: Thermal Control Subsystem Requirements Verification Plans Table

Req #	Requirement	Verification method	Rationale for Method	Preliminary Verification Plan
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TH-2.1	The Thermal subsystem shall maintain a system equilibrium temperature of 370 K.	Test	The Thermal Control System must be tested to ensure it can effectively work as a design in the system-level	The Aerobot system is designed to withstand temperatures slightly above 370 K and so thus it can be heated up and the temperatures can be monitored to see if the Thermal Control System works effectively.
TH-2.2	All Thermal subsystem components shall optimally perform in the temperature range of 350 K - 400 K.	Test	The Thermal Control System components are capable of withstanding Venus-like temperature of 350 K -400 K and so thus can be tested on site if they perform effectively.	The Thermal Control System can be observed during testing of the Aerobot system as a whole by monitoring each components performance and contribution.
TH-2.3	All Aerobot system chemical coating and white paint surfaces shall perform optimally.	Test	The chemical coating and white paint are responsible for radiating out a bulk of the heat and the best to make sure is by testing if performance and contribution is as expected.	The chemical coating and white paint can be tested either on individual surfaces after manufacturing or on the Aerobot as a whole during testing of the radiator to monitor performance.
TH-2.4	The Aerobot system shall be equipped with an optimally functioning	Test	The ACT radiator is the most important and capable part of the Thermal Control System	The performance of the radiator must be carefully observed individually at different temperature differences to measure its robustness.

	ACT radiator to provide with the needed amount of 30 kW cooling in the hottest and 11 kW of heating in the coldest conditions.		and must be able to provide effective cooling and trap heat. Thus it is crucial to test its performance in both the coldest and hottest conditions.	
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2.1.5. Payload Subsystem Overview

Science instruments that are included on team Igneous Seven's vehicle for the Aerobot mission are the Dust Flux Monitor Instrument (DFMI), Spectroscopy for Investigation of Characteristics of the Atmosphere of Venus (SPICAV-IR), and Venus Interferometric Synthetic Aperture Radar (VISAR).

How the DFMI operates, According to NASA Jet Propulsion Laboratory, "*Dust Flux Monitor Instrument (DFMI)*", states, "...The Dust Flux Monitor Subsystem includes a dedicated, sophisticated Dust Flux Monitor Instrument (DFMI) for small particles, and two separate acoustic impact sensors that monitor for strikes by the relatively rare, but especially dangerous larger particles." ("Stardust | JPL | NASA," n.d.). DFMI detects small particles and uses acoustics for the occasional larger particle that may be detected by the system. Technology readiness level for the "Dust Flux Monitor Instrument" is level 7. The Dust Flux Monitor was designed to detect dust particles as small as only 6 microns thick, but it can detect larger particles as well. This means that the Dust Flux Monitor should be capable of fulfilling the objective of measuring the mass flux of plume particles. It should be noted that the Dust Flux monitor was not used on Venus, but was used on several missions such as the Soviet Union's Vega missions to comet Haley, and the Cassini mission to Saturn in 1997. Since the instrument was shown to operate within a planetary environment, and was able to accomplish a similar goal to our objective, but was not